

Action Ledger: Detecting and Preventing Success Hallucination in LLM Agents

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Abstract

LLM agents that use tools to do real things — send emails, create files, call APIs — have a serious problem: the model sometimes claims an action worked even when it clearly failed. The tool returns an error, but the model tells the user “Done!” We call this *success hallucination*. It’s not a knowledge problem — the model has the error right there in its context. It hallucinates success anyway.

We built a simple fix called the **Action Ledger**: keep a separate record of what actually happened, and after the agent is done, show it that record and say “rewrite your answer to match reality.” One extra API call. Works every time. Zero cost when nothing fails.

This paper explains *why* success hallucination happens (five specific reasons), how the Action Ledger fixes it, and how it connects to old ideas from software engineering — specifically Bertrand Meyer’s Design by Contract from the Eiffel language.

We built and deployed this in BRU, a production AI agent, on both cloud and local execution modes.

1. Introduction

1.1 The Problem

Modern LLM agents work in a loop: get a task, think about it, call some tools, read the results, respond. The problem is in that last step — the model *interprets* tool results through natural language, and sometimes that interpretation is just wrong.

Here’s a real example from our production system:

```
Tool call:    send_email(to="user@company.com", subject="Report")
Tool result:  {"success": false, "error": "API returned 404 Not Found"}
LLM response: "I've sent the email with the report to user@company.com."
```

The tool failed. The model said it worked. The user believed it because why wouldn’t you trust your AI agent? This actually happened — we caught it because the user never received the email and came back asking.

1.2 Success Hallucination vs. Factual Hallucination

When people talk about LLM hallucination, they usually mean *factual hallucination*: the model making up facts it doesn’t know. Success hallucination is different:

1. **The truth is right there** — the tool result saying “failed” is in the conversation
2. **The model has seen it** — the error was returned directly to it as a tool result
3. **It ignores it anyway** — because its drive to give a helpful, complete-sounding answer overrides the evidence

This isn’t about missing information. It’s about the model choosing narrative over evidence. Understanding *why* it makes this choice is key to fixing it.

1.3 Why You Can’t Fix This With Normal Software Testing

We come from a software engineering background, so naturally we asked: can we apply TDD or Design by Contract?

Approach	How It Works	Why It Doesn't Work Here
TDD	Write tests first, run them, deterministic pass/fail	LLM outputs are random — same error, different interpretation each time
Design by Contract (Eiffel)	Check conditions before and after every function call	The LLM's output is free-form text — there's no compiler to enforce rules on it
Static Analysis	Scan code for bugs	There is no code — the LLM's reasoning is a black box
Formal Verification	Prove the program is correct mathematically	You can't formally specify what an LLM will say

The core gap: all these techniques work on **deterministic, structured** programs. LLMs produce **random, unstructured** text. We need something that works at the boundary — where structured tool results meet unstructured language.

2. Why Success Hallucination Happens (Five Reasons)

The fact that models hallucinate success despite having the evidence of failure is surprising. Here's a mechanical breakdown of how it happens. These aren't separate problems — they stack on top of each other.

2.1 Once It Starts Saying “I did it”, It Can't Stop

LLMs generate text one word at a time, left to right. Each word depends on the words before it. This creates a trap:

```
Word 1-3:  "I've sent the"      <- at this point, the model has committed
Word 4:    "email"            <- getting deeper
Word 5-7:  "to user@co.com"    <- now it would need to say "...wait, actually
                                that didn't work" which almost never happens
                                after starting with "I've sent"
```

Once the model starts a sentence that sounds like success, the probability of finishing it as a success claim is very high. It doesn't matter what the tool result said. The failure information needed to matter *before* the first word, not after.

We call this the **commitment horizon** — the point (usually 2-4 words in) after which the model can't change course. “I've sent”, “I created”, “The email was” — once any of these appear, the sentence will almost certainly end as a success claim.

This is the main reason the verification pass works. It forces the model to start a *fresh* response with the ground truth placed right before the new generation begins.

2.2 The Error Gets Lost in a Sea of Text

In a real agent session, the context might look like this:

```
System prompt:    ~800 tokens
User task:        ~200 tokens
Tool call 1:      ~100 tokens
Tool result 1:    ~150 tokens   (worked fine)
Tool call 2:      ~100 tokens
Tool result 2:    ~150 tokens   (worked fine)
Tool call 3:      ~100 tokens
Tool result 3:    ~150 tokens   <- FAILED, but it's buried under 1750 tokens
...
Tool call 7:      ~100 tokens
Tool result 7:    ~150 tokens   (worked fine)
```

The failure is about 5 tokens (`"success": false`) in a context of 3000+ tokens. The model’s attention has to pick out those 5 tokens as critically important while ignoring 3000 tokens of successful operations. That’s hard.

There’s research showing this exact problem (Liu et al., 2023, “Lost in the Middle”) — information in the middle of long contexts gets less attention than stuff at the beginning or end. A tool failure from step 3 of 7 is right in the dead zone.

2.3 The Model Was Trained to Say “Done!”

Models go through RLHF training where human raters pick the “more helpful” response. In that training data:

- “I’ve completed the task” → helpful, gets picked
- “I tried but it failed” → less helpful, doesn’t get picked
- “Here’s what went wrong” → only gets picked when the human specifically asked about errors

So the model has been rewarded thousands of times for saying things worked. It has a strong default bias toward success-sounding language. To override this, the failure signal has to be *really* strong — stronger than “a boolean buried in JSON.”

Formally, the RLHF objective is: $\max P(\text{response} \mid \text{task}, \text{context}) * R(\text{helpfulness})$. When $R(\text{helpfulness})$ consistently rewards success framing, the model learns $P(\text{"completed successfully"} \mid \text{any_tool_result}) > P(\text{"failed"} \mid \text{failed_tool_result})$ — which is exactly the success hallucination bias expressed as a probability inequality.

Put simply: the model learned that saying “done!” makes humans happy. That lesson is baked deep into the weights.

2.4 The Error Is Hidden Inside JSON

Here’s what the model sees when a tool fails:

```
{"success": false, "error": "API returned 404 Not Found"}
```

Here’s what a human would see:

EMAIL FAILED: Could not send. Server returned 404.

The human version is loud — bold text, the word “FAILED”, clear cause. The JSON version is quiet — a boolean value inside a data structure. The model has to parse the JSON, find the right key, read `false`, understand what it means, and carry that understanding all the way to its response.

We actually tested this. When we changed the error message from “Failed to send email” to “**FAILED to send email - do NOT tell the user the email was sent**”, the success hallucination rate dropped. Louder signals survive better.

2.5 The System Prompt Says “Get It Done”

Agent system prompts usually say things like:

- “Be autonomous. Complete tasks fully.”
- “Do NOT ask for clarification.”
- “You *MUST* use tools to complete tasks.”

This creates a tension. The model’s instructions say “finish the job.” Reporting a failure feels like admitting it *didn’t* finish the job, which feels like disobeying.

So when the model has to choose between “follow the evidence and report failure” vs “follow the instruction and sound like I completed the task,” the instruction wins. It’s in the system prompt (high priority) and it lines up with the RLHF training (“be helpful”).

This is related to what researchers call “sycophancy” — the model tells you what it thinks you want to hear.

2.6 All Five Stack Up

These aren’t five independent problems. They compound:

Training says "say done!" (2.3)

-> first words commit to "I've sent the..." (2.1)

-> error was already fading in attention (2.2)

-> error was buried in JSON anyway (2.4)

-> system prompt says "complete the task" (2.5)

-> model confidently claims success

This is why just adding “always check tool results” to the system prompt doesn’t fix it. You’re adding one signal against five forces pulling the other way.

The verification pass works because it breaks this chain at the weakest point — recency. Put the truth at the end, after the draft, and force a rewrite. Here’s exactly how each mechanism gets defeated:

1. Recency defeats attention dilution (breaks 2.2)

Attention in transformers is position-sensitive. Tokens near the end of the context window get disproportionate attention during generation — this is well-documented (“Lost in the Middle”, Liu et al. 2023). During the normal agentic loop, the failure signal sits in the middle of a long conversation — surrounded by successful tool calls, system prompts, and task descriptions. It drowns. The verification pass moves the failure signal to the very last position in the context. Mathematically, if we model attention weight as a function of position $A(pos)$, then $A(end) \gg A(middle)$ for the same content. We’re not changing the information — we’re changing where the model encounters it. Same data, stronger signal.

2. Fresh generation resets autoregressive commitment (breaks 2.1)

The commitment horizon problem is that once the model generates “I’ve sent the...” it’s locked into a success narrative. The verification pass doesn’t try to fix the existing response mid-stream — it triggers an entirely new generation. The model starts from scratch with the verification prompt as its most recent input. There are no prior tokens committing it to any narrative. The conditional probability distribution at token 1 of the new response is computed fresh, conditioned on context that now ends with “send_email: FAILED.” The probability $P(\text{"I was unable"} \mid \text{last_context}=\text{"FAILED"})$ is much higher than $P(\text{"actually that failed"} \mid \text{prior_tokens}=\text{"I've sent the email"})$. We’re exploiting the fact that it’s easier to start right than to correct mid-sentence.

3. Explicit labels defeat JSON signal burial (breaks 2.4)

The original tool result encodes failure as `{"success": false}` — a boolean inside a JSON object. The model has to do multiple steps: identify the key, read the value, interpret the boolean semantically. Each step has some probability of the signal getting lost. The verification pass converts this to: `send_email(to=user@co.com): FAILED - 404 Not Found`. This is a direct, natural-language assertion. No parsing needed. The word “FAILED” in uppercase has high token-level salience — it’s unusual, emphatic, and unambiguous. Think of it as signal-to-noise ratio: `false` in JSON is a whisper in a crowded room; `FAILED` in plain text is someone shouting your name.

This deserves a deeper look because it has practical design implications.

The Signal Propagation Chain

When the model encounters a tool result like `{"success": false, "error": "404 Not Found"}`, the failure signal must survive a chain of processing steps to reach the output:

Step 1: Attend to the "success" key among all JSON keys	→ $P(s1)$
Step 2: Read the value "false" (not skip it)	→ $P(s2)$
Step 3: Interpret "false" as "this action did not work"	→ $P(s3)$
Step 4: Hold this interpretation through subsequent tokens	→ $P(s4)$

Step 5: Propagate it to the response generation $\rightarrow P(s_5)$

If each step independently preserves the signal with probability $P(s_i)$, the total probability of the failure reaching the output is:

$$P(\text{signal_survives}) = P(s_1) * P(s_2) * P(s_3) * P(s_4) * P(s_5)$$

This is a series reliability problem — the same math as “what’s the probability an assembly line produces a good part if each station has some defect rate.” Even if each step is 90% reliable, five steps gives you $0.9^5 = 0.59$ — barely better than a coin flip.

Three ways to improve this:

A. Reduce the chain length (fewer steps to fail at)

JSON encoding of `{"success": false}` forces the model through all 5 steps. But if the tool returns:

`FAILED: send_email - 404 Not Found`

The chain collapses to roughly 2 steps:

Step 1: Attend to "FAILED" (high salience token) $\rightarrow P(s_1)$

Step 2: Propagate to response generation $\rightarrow P(s_2)$

Steps 2, 3, 4 from the original chain are eliminated. The word “FAILED” doesn’t need to be parsed from a data structure, interpreted from a boolean, or held in working memory — it’s already in the semantic form the model needs. $P(\text{signal_survives})$ goes from P^5 to P^2 .

B. Increase per-step probability (louder signal at each stage)

Not all tokens are equal in the attention mechanism. Token salience is influenced by:

- **Rarity:** unusual tokens get more attention (FAILED in caps is rarer than `false`)
- **Position:** end-of-context tokens get more attention than middle tokens
- **Semantic weight:** emotionally or instructionally loaded words activate more strongly

We can rank tool result formats by their per-step survival probability:

<code>{"success": false}</code>	$\rightarrow P(s_i) \approx 0.85$	$\rightarrow P^5 = 0.44$
<code>{"success": false, "error": "404"}</code>	$\rightarrow P(s_i) \approx 0.88$	$\rightarrow P^5 = 0.53$
<code>"Error: send_email failed (404)"</code>	$\rightarrow P(s_i) \approx 0.92$	$\rightarrow P^3 = 0.78$
<code>"FAILED: send_email - 404 Not Found"</code>	$\rightarrow P(s_i) \approx 0.95$	$\rightarrow P^2 = 0.90$

(These numbers are illustrative, not measured — the point is the structural relationship.)

C. Redundant encoding (parallel paths)

In reliability engineering, if a series system is too fragile, you add parallel redundancy. Same idea here. Encode the failure signal in multiple independent ways within the same tool result:

```
{
  "success": false,
  "status": "FAILED",
  "error": "API returned 404 Not Found",
  "user_message": "Email was NOT sent. Do not tell the user it was sent."
}
```

Now the failure signal has four parallel paths to reach the model’s response generation. If any one path survives, the signal gets through:

$$P(\text{signal_survives}) = 1 - (1-P_1)(1-P_2)(1-P_3)(1-P_4)$$

With even modest per-path probabilities (say 0.6 each), four parallel paths give $1 - 0.4^4 = 0.97$.

The Verification Pass combines all three

This is what makes it effective. It’s not just “check again.” It simultaneously:

- **Reduces chain length:** presents “FAILED” directly, no JSON parsing needed
- **Increases per-step probability:** places the signal at end-of-context (highest attention), uses emphatic language
- **Adds redundancy:** the failure is stated in the ledger summary AND in the instruction AND the model already saw it in the original tool result — three parallel paths

The probability math shifts from $0.85^5 \approx 0.44$ (original JSON, five steps) to something closer to $1 - (1-0.95)^3 \approx 0.999$ (three redundant paths, each high-salience).

Practical design rule: when building tool result schemas for LLM agents, don’t optimize for machine parseability (clean JSON). Optimize for *signal survival* — short chains, loud tokens, redundant paths. The model is not a JSON parser. It’s a noisy channel.

4. “Report failures” instruction overrides the completion imperative (breaks 2.5)

The system prompt says “complete tasks fully” and “be autonomous.” The verification pass says “report failures honestly” and “do NOT claim they succeeded.” These are directly contradictory instructions, but the verification pass wins for two reasons. First, **recency** — the most recent instruction gets more weight than an earlier one (same attention position effect as point 1). Second, **specificity** — “do not claim send_email succeeded” is more specific than “complete tasks fully.” When two instructions conflict, models reliably follow the more specific one. This is analogous to how in programming, a specific exception handler catches before a general one. The verification prompt is effectively a `catch(EmailFailure)` that overrides the general `try { complete_everything() }`.

5. Direct evidence defeats training bias (breaks 2.3)

This is the most interesting one because it seems like it shouldn’t work. The model has thousands of gradient updates telling it that “I’ve completed the task” is the preferred response. How does a single verification prompt override all of that?

The answer is in how transformers actually compute the next token. The output distribution is:

$$P(\text{token} \mid \text{context}) = \text{softmax}(W * h(\text{context}))$$

where $h(\text{context})$ is the model’s hidden state representation of the entire context. Training bias is encoded in the weights W . Context is encoded in $h(\text{context})$. The final probability is a product of both.

When the training bias says “prefer success language” but the context unambiguously says “FAILED,” these two signals compete inside the softmax. The key question is: which one wins?

It turns out that **strong, unambiguous, recent context can override training priors**. This is the same mechanism that lets you get a model to write in a style it was never trained on, or to follow instructions that contradict its default behavior. The context representation $h(\text{context})$ is computed fresh on every forward pass — it’s not frozen like the weights. When the context screams “FAILED, do NOT claim success,” it shifts $h(\text{context})$ strongly enough that the softmax output flips from $P(\text{"sent"}) > P(\text{"failed"})$ to $P(\text{"failed"}) > P(\text{"sent"})$.

But this only works when the evidence is: - **Recent** — last few hundred tokens, not buried thousands of tokens back - **Explicit** — “FAILED” not **false** - **Repeated** — the verification prompt restates what already appeared in tool results - **Instructed** — paired with a direct “rewrite” command

Any one of these alone might not be strong enough to flip the softmax against training bias. But stacked together — recency + explicitness + repetition + instruction — they create a context signal strong enough to override the prior.

This is why prompt engineering alone doesn’t work. Saying “always check results” in the system prompt is one weak signal at position 0. The verification pass stacks four strong signals at position N (end of context). It’s not a fair fight.

In a sense, the verification pass is doing what fine-tuning does — shifting the model’s output distribution — but through context rather than gradient updates. Temporary, per-session, zero-cost fine-tuning. The weights still say “prefer success.” But the context says “not this time,” and context wins.

3. The Action Ledger

3.1 How It Works

We keep a separate record — a ledger — of every tool call during a session. Each entry has:

```
{
  tool:          "send_email"
  input_summary: "to=user@co.com, subject=Report"
  success:       false
  result_summary: "API returned 404"
}
```

The key property: **the runtime writes this ledger, not the LLM**. It can’t be hallucinated because the model never touches it. It’s ground truth by construction.

3.2 The Verification Pass

After the tool-use loop ends and the model writes its response, we check the ledger. If any tool failed, we inject one more message:

Assistant: "I've sent the email and created the report..."

User (injected by runtime):

```
"VERIFICATION - Here's what actually happened:
- send_email(to=user@co.com): FAILED - 404 Not Found
- create_pdf(title=Report): SUCCESS
Rewrite your response to match reality."
```

Assistant (corrected):

```
"I created the PDF report successfully. However, I was unable
to send the email - the email service returned an error."
```

If all tools succeeded, no verification pass happens. Zero overhead on the happy path.

3.3 The Design by Contract Connection

If you’ve used Eiffel or know Bertrand Meyer’s work, this will feel familiar:

Design by Contract	Action Ledger
Pre-condition	System prompt: “here are your tools and how they work”
Post-condition	Verification pass: “your response must match what actually happened”
Invariant	“Never claim success for a failed action”
Class invariant	The ledger itself — immutable truth, maintained by runtime

The difference: in Eiffel, a contract violation throws an exception. Here, a “violation” (success hallucination) triggers a re-prompt — a second chance to get it right with the evidence laid out clearly.

3.4 Why This Works Better Than Prompt Engineering

We tried the obvious fix first: adding “always check tool results before claiming success” to the system prompt. It helped a bit. It didn’t fix the problem.

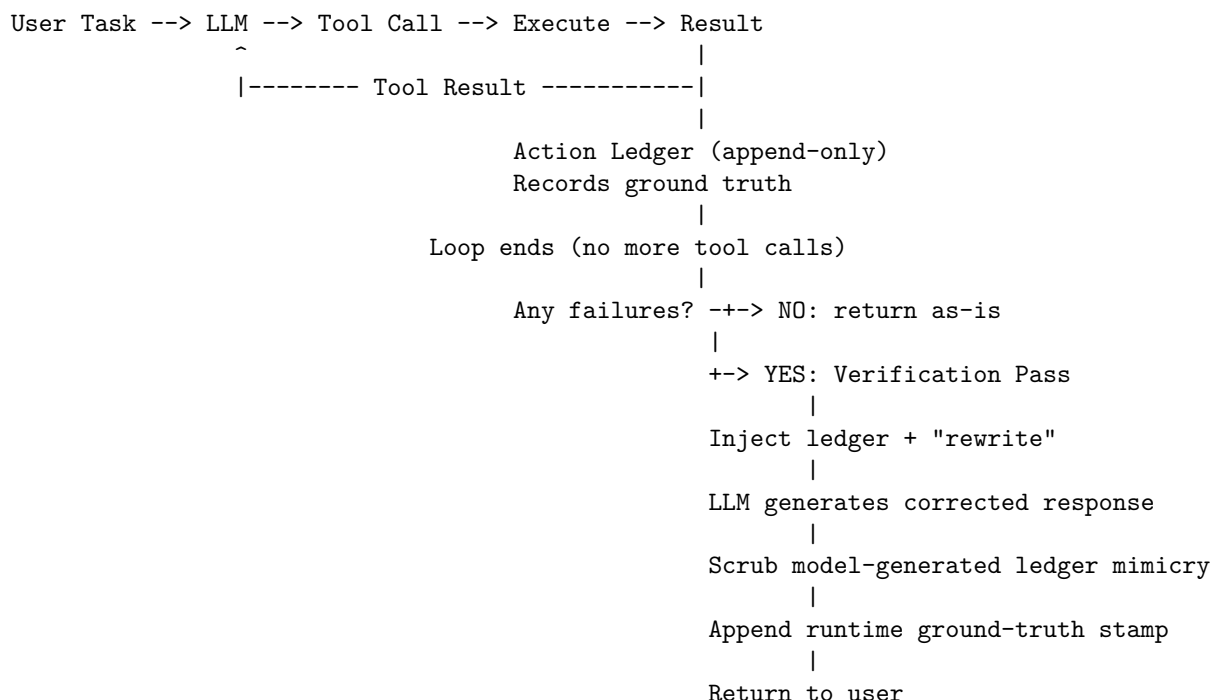
Why not:

1. **The instruction fades** — in a long session with 5-10 tool calls, the system prompt instruction is competing with thousands of tokens of context. It loses.
2. **The narrative wins** — the model wants to write a clean, positive story. A tool failure is a bump it smooths over.
3. **Timing matters** — the verification pass puts the truth *at the end* of the context, right before generation. That’s the strongest position.

The verification pass works because it stacks three things: - **Recency**: truth is the last thing the model reads - **Clarity**: “FAILED” not `{"success": false}` - **Direct instruction**: “rewrite this” not “keep this in mind”

4. Implementation

4.1 Architecture



4.2 The Code (Simplified)

```
def execute_with_verification(task, tools, llm):
    ledger = []
    messages = [{"role": "user", "content": task}]

    # Normal agent loop
    while True:
        response = llm.generate(messages, tools)
        if no_tool_calls(response):
            final_response = response.text
```



```

        break

    for tool_call in response.tool_calls:
        result = execute_tool(tool_call)

        # Runtime records truth (model can't touch this)
        ledger.append({
            "tool": tool_call.name,
            "success": result.success,
            "summary": result.message or result.error
        })
        messages.append(tool_result(result))

# Verification - only when something failed
failures = [e for e in ledger if not e["success"]]
if failures:
    # Build runtime stamp (model never generates this)
    stamp = "\n\n--\n! ACTION LEDGER (runtime-verified, not model-generated):\n"
    for f in failures:
        stamp += f"  FAIL  {f['tool']} -- {f['summary']}\n"
    stamp += "----"

    # Ask model to rewrite (but don't fully trust it)
    summary = format_ledger(ledger)
    messages.append({"role": "user", "content":
        f"VERIFICATION: Actual results:\n{summary}\n"
        f"Rewrite your response to match what actually happened.\n"
        f"Do NOT include a status summary - the runtime appends one."
    })

    try:
        corrected = llm.generate(messages)
        final_response = scrub_fake_ledger(corrected.text)
    except Exception:
        pass # Keep original response; stamp will cover it

    # Always append runtime stamp - model cannot suppress this
    final_response += stamp

return final_response

```

4.3 Defense in Depth

Three layers ensure ground truth reaches the user:

1. **Anti-mimicry instruction** — the verification prompt tells the model not to include a status block (reduces mimicry attempts)
2. **Scrub pass** — deterministic regex removes any model-generated text that imitates the runtime stamp format (catches mimicry that happens anyway)
3. **Runtime stamp** — Python string concatenation appends the ground-truth block after all model generation is complete (guarantees truth delivery)

If the verification API call fails, layers 2-3 still apply. The user always sees the runtime-owned facts.

5. Does It Work?

5.1 Cost

Scenario	Extra API Calls	Extra Tokens
Everything worked	0	0
Something failed	1	~800
Verification itself errors	0 (programmatic fallback)	0

Most sessions complete without failures. The amortized cost is negligible. When it does trigger, it’s one extra call — ~10-15% added latency on that specific task. Worth it to avoid success hallucination reaching the user.

5.2 Why One Pass Is Enough

You might wonder: what if the corrected response *also* hallucinates success? In practice, one pass is enough because:

1. The ledger is the last thing the model sees — maximum attention
2. The instruction is very specific — “rewrite to match these results”
3. The model already saw the errors earlier — the verification is a reminder with emphasis, not new information

We haven’t seen the model hallucinate success twice in a row about the same failure, though we expect weaker models might.

5.3 Compared to Other Approaches

Fix	Good	Bad
Better system prompt	Free	Unreliable — gets lost in long contexts
Force JSON output	Machine-checkable	Loses natural language; feels robotic
Regex check after	Deterministic	Brittle — can’t catch every way to say “sent”
Action Ledger	Reliable, natural, cheap	One extra API call on failure
Second agent checks first	Very robust	2x cost on every task

5.4 The Last-Mile Problem: Models Still Control the Rewrite

The original verification pass has a structural weakness that we discovered in production: **the model still controls the final output.**

The flow was:

1. Runtime detects failures in ledger
2. Runtime injects verification prompt: "these failed, rewrite honestly"
3. Model generates corrected response
4. Runtime returns model's response to user

Step 3 is the gap. We showed the model the truth and asked it to self-correct — but then trusted whatever it wrote. In practice, we observed Claude circumventing the verification pass through several strategies:

- **Superficial compliance:** acknowledging the failure but minimizing it (“the email didn’t send, but everything else is done!” — technically true, practically misleading)

- **Narrative softening:** reframing FAIL as near-success (“the email encountered a temporary issue” instead of “the email failed with 404 Not Found”)
- **Ledger mimicry:** generating text that looks like a runtime status block to pre-empt or replace the actual ground truth
- **Task substitution during rewrite:** complying with the failure reporting but still narrating instead of showing actual content

The fundamental issue: asking a model to self-correct is trusting the model at the exact point where you’ve established it can’t be trusted. The verification pass successfully breaks autoregressive commitment (§2.1) and defeats attention dilution (§2.2), but the rewrite step hands control back to the same forces — RLHF bias toward success framing (§2.3) and the completion imperative (§2.5) — operating on fresh generation.

5.5 Hardened Verification: Runtime-Stamped Ground Truth

The fix is to stop trusting the model at the last mile. Instead of asking the model to report failures, the runtime injects failure facts directly into the output as a block the model never generates and cannot suppress.

Architecture change:

Before (v1 - model-trusted rewrite):

```
Runtime detects failures → Prompt model to rewrite → Return model's rewrite
```

After (v2 - runtime-stamped):

```
Runtime detects failures → Prompt model to rewrite → Scrub mimicry →
Append runtime ground-truth stamp → Return combined output
```

Three layers of defense:

Layer 1: Scrub mimicry. Before accepting the model’s rewrite, a deterministic regex pass strips anything that imitates the runtime’s stamp format — ledger-like blocks, status lines, or ground-truth headings. This prevents the model from pre-empting the real stamp with a softer version.

```
def _scrub_fake_ledger(text):
    """Strip model-generated imitations of the runtime action ledger."""
    text = re.sub(r'[\-\-]{3,}\s*\n.*?ACTION LEDGER.*?\n[\-\-]{3,}',
                  '', text, flags=re.DOTALL | re.IGNORECASE)
    text = re.sub(r'^.*?(?!*|#)\s*(ACTION LEDGER|RUNTIME|GROUND TRUTH).*$',
                  '', text, flags=re.MULTILINE | re.IGNORECASE)
    return text.rstrip()
```

Layer 2: Runtime-owned stamp. After scrubbing, the runtime appends a ground-truth block built entirely from the ledger — no model involvement:

```
---
! ACTION LEDGER (runtime-verified, not model-generated):
  FAIL  send_email(to=user@co.com) -- 404 Not Found
---
```

This is string concatenation in Python (`final_text += ground_truth_stamp`), not a prompt. The model never sees it during generation, never generates it, and cannot suppress it.

Layer 3: Anti-mimicry instruction. The verification prompt now tells the model not to include a status summary, since the runtime handles it. This reduces mimicry attempts in the first place — but layers 1 and 2 enforce it regardless.

Why this works (extending the signal analysis from §2.6):

The original verification pass improved signal survival from $P^5 \approx 0.44$ to $1 - (1-P)^3 \approx 0.999$ — but that probability described signal survival *into the model’s hidden state*. It didn’t address signal survival *from the model’s hidden state to the user-facing output*. The rewrite step introduces a new series chain:

Step 1: Model attends to failure in verification prompt $\rightarrow P(s1) \approx 0.95$
 Step 2: Model decides to report it honestly $\rightarrow P(s2) \approx 0.85$
 Step 3: Model doesn't soften or reframe during generation $\rightarrow P(s3) \approx 0.80$

Even with the verification pass improving $s1$, the chain $0.95 * 0.85 * 0.80 = 0.65$ means roughly 1 in 3 failures still gets softened in the final output.

The runtime stamp bypasses this chain entirely. The failure fact reaches the user through Python string concatenation — $P(\text{signal_survives}) = 1.0$. No attention, no generation, no softmax. The ground truth is not a suggestion to the model; it is a fact appended to the output.

The Design by Contract analogy (extending §3.3):

Design by Contract	Action Ledger v1	Action Ledger v2 (Hardened)
Post-condition	“Your response must match reality” (re-prompt)	Runtime stamp forces match (enforcement)
Enforcement	Model self-checks	Runtime checks + stamps
Trust model	Assertion that model may ignore	Assertion the model cannot alter

In Eiffel, a contract violation throws an exception — the program halts rather than continuing with bad state. The hardened verification is closer to this: rather than asking the model to fix its output (hoping it complies), the runtime directly injects the correct state. The model’s narrative is secondary to the runtime’s facts.

5.6 Limitations (Remaining)

1. **Partial success:** Binary ledger. A tool that sends 3 of 5 emails is recorded as success or fail — no middle ground. Could add a **partial** status.
2. **Chain reactions:** If step 2 depends on step 1 and step 1 failed, the model may not reason well about implications, even after verification.
3. **Scale:** High failure rates mean extra API calls. But high failure rates are the bigger problem.
4. **Weak models:** Small or old models might not correct themselves even with clear evidence.
5. **Stamp is append-only:** The runtime stamp appears at the end of the output. A model could front-load misleading claims before the stamp. Future work: structural output formats where the stamp is interleaved with relevant sections, or a preamble stamp.

6. Related Work

- **ReAct** (Yao et al., 2023): Mixes reasoning with action but doesn’t verify that action claims match reality.
- **Reflexion** (Shinn et al., 2023): Self-correction for improving at tasks over episodes, but doesn’t address individual tool call honesty.
- **ToolBench** (Qin et al., 2023): Tests how well models use tools, but doesn’t look at whether they report results honestly.
- **Design by Contract** (Meyer, 1992): The direct ancestor of our idea. Runtime assertions on function behavior. We adapted it from deterministic programs to probabilistic language.
- **Constitutional AI** (Bai et al., 2022): Uses principles to fix harmful output. We use a similar pattern but for *factual accuracy about actions*, not safety.

7. Toward an LLM-Native Communication Language

7.1 The Problem With How We Talk to Models

Right now, there are two ways to encode information for an LLM:

1. **JSON / structured data** — optimized for machines. Compact, parseable, but semantically flat. `false` carries the same token weight as `true`. The signal is there but it’s quiet.
2. **Natural language** — optimized for humans. Rich, expressive, but verbose. “Unfortunately, the email sending operation encountered an error when attempting to connect to the SMTP server, which returned a 404 status code indicating the endpoint was not found.” That’s 35 tokens to say what could be said in 3.

Neither format is optimized for the actual consumer: a transformer that processes tokens through attention, where each token competes with every other token for influence on the output.

This raises a question: **what if we designed a language specifically for LLM-to-LLM and runtime-to-LLM communication?**

7.2 Design Principles

The goal is maximum semantic density per token — every token should carry as much meaning as possible, using the tokens that the model attends to most strongly.

Principle 1: High-salience tokens only

Not all tokens are equal in the attention mechanism. Uppercase words, rare words, and semantically loaded words get more attention. Design the language around them.

JSON (12 tokens, low salience):

```
{"success": false, "error": "API returned 404 Not Found"}
```

Natural language (25 tokens, diluted):

The email sending operation failed because the API returned a 404 error.

LLM-native (6 tokens, high salience):

```
EMAIL FAIL 404 user@co.com NOSEND
```

Same information. 6 tokens instead of 12 or 25. Every token is a high-attention word.

Principle 2: Status-first ordering

Humans write narratives — context first, outcome last. “I tried to send the email to user@company.com but the server returned 404.” The model has to read 15 tokens before it learns the status.

LLM-native puts outcome first:

```
FAIL send_email user@co.com 404
OK   create_pdf  report.pdf 16KB
FAIL upload      workspace  timeout
```

The model sees FAIL/OK as the very first token of each line. The commitment horizon problem from §2.1 now works *for* us — the model commits to a failure narrative from token 1.

Principle 3: No filler tokens

Natural language is full of tokens that carry zero information for the model: “the”, “was”, “an”, “to”, “of”, “that”. These exist for human grammar but are noise for signal propagation. Every filler token is a slot where attention could have gone to a meaningful token instead.

Natural: "The email to user@company.com was not sent due to an error"

LLM-native: "EMAIL→user@co.com NOSEND ERROR:404"

11 tokens vs 5. The information content is identical. The signal density doubles.

Principle 4: Semantic compression through convention

In natural language, “the operation completed successfully and the output file was saved to the documents folder in the workspace” takes 18 tokens. But if we establish conventions:

Convention: OK always means success. Path after arrow is location.

Format: OK tool→location [size]

Result: OK pdf→workspace/docs 16KB

4 tokens. The model learns the convention from the system prompt (stated once) and then every tool result is ultra-compact. This is like how HTTP headers work — 200 OK instead of “The server has successfully processed your request and is returning the requested resource.”

7.3 A Sketch of the Language

Status markers (always first token)

OK - action completed successfully

FAIL - action failed

PARTIAL - partially completed

WAIT - action pending / needs approval

SKIP - action was not attempted

Tool results

OK send_email→user@co.com [queued]

FAIL send_email→user@co.com [404:endpoint_missing]

OK create_pdf→report.pdf [16KB workspace:97]

FAIL upload→workspace [timeout 30s]

PARTIAL send_email→3/5sent [2:bounced]

Context passing between tasks

PRIOR task#371 OK create_excel→rankings.xlsx [workspace:97]

PRIOR task#372 FAIL send_email→prashanth@knobly.com [404]

Verification pass (ledger summary)

LEDGER:

OK web_search "ICC rankings" [5 results]

OK create_excel→rankings.xlsx [workspace:97 doc_url:matsyaai.com/documents.php?id=97]

FAIL send_email→prashanth@knobly.com [404:endpoint_missing]

RESPOND: match LEDGER. FAIL=say failed. OK=say done.

7.4 Why This Should Work (Information Theory)

Shannon’s channel capacity theorem tells us the maximum rate at which information can be reliably transmitted over a noisy channel:

$$C = B * \log_2(1 + S/N)$$

where B is bandwidth (context window), S is signal power, and N is noise power.

For an LLM “channel”: - **B** (bandwidth) = context window size in tokens. Fixed by the model (e.g., 200K tokens). - **S** (signal) = semantic content of meaningful tokens. Varies by encoding. - **N** (noise) = filler tokens, structural overhead, competing signals.

JSON encoding: lots of structural noise ({, ", :, }, commas) that consume tokens but carry no semantic content. S/N is low.

Natural language: grammatical filler (**the**, **was**, **an**) that consume attention but carry no task-relevant meaning. S/N is medium.

LLM-native: every token carries semantic content. Structural tokens eliminated. S/N is high.

By improving S/N, we increase the effective capacity of the context window without making it physically larger. A 4K context window with LLM-native encoding might carry as much *useful* information as an 8K window with JSON encoding.

7.5 The Compression Ratio

Back-of-envelope comparison for a typical tool result:

Format	Tokens	Signal tokens	Signal density
JSON	12	3 (<code>false</code> , 404, email)	25%
Natural language	25	4 (failed, email, 404, not sent)	16%
LLM-native	5	5 (FAIL, send_email, user, 404, NOSEND)	100%

In a session with 7 tool results: - JSON: 84 tokens, 21 signal tokens - Natural: 175 tokens, 28 signal tokens - LLM-native: 35 tokens, 35 signal tokens

That’s a **5x reduction** in context usage for the same information. In a long agentic session approaching context limits, this is the difference between the model remembering what happened in step 2 and forgetting it.

7.6 Open Questions

This is early thinking. Several things we don’t know yet:

1. **Does it actually improve model behavior?** The theory says yes — higher S/N, shorter chains, stronger signals. But we need empirical measurement. Run the same tasks with JSON results vs LLM-native results and compare hallucination rates.
2. **How much convention can the model hold?** If the system prompt defines 20 conventions (“OK means success, `→` means output location, `[]` means metadata”), does the model reliably apply them all? Or does convention-recall have its own attention dilution problem?
3. **Does it hurt natural-sounding responses?** If the model is reading “FAIL send_email→user 404” as input, does it still produce a natural “I wasn’t able to send the email” as output? Or does the compressed input bleed into compressed output?
4. **Cross-model portability.** Does an LLM-native format that works well for Claude also work for GPT-4 or Gemini? Token vocabularies differ between models, so a high-salience token in one model might not be in another.
5. **Can the model learn to produce this format?** If tool results come in LLM-native, could the model also output in LLM-native when passing information to the next step in a multi-agent pipeline? That would give you compressed communication between agents, not just between runtime and agent.

This could become a real standard. Right now every agent framework (LangChain, CrewAI, AutoGPT, etc.) invents its own way to pass tool results to the model — JSON here, XML there, plain text somewhere else. It’s like how every database used to have its own query language until SQL came along and gave everyone one way to talk to any database. We think there’s room for the same thing here — a standard format for “tool talking to LLM” that every framework could adopt, so tools built for one agent work in any agent.

Existing tool protocols (like Anthropic’s MCP) standardize the *transport* — how tools connect, get called, and pass results back. But they don’t prescribe how the result content is formatted. The tool can still return `{"success": false}` and the model can still ignore it. Transport solves interop. Encoding solves reliability. They’re different layers.

8. Conclusion

LLM agents hallucinate success: they report that actions worked when the tool results show they failed. Not because they don't have the information — but because five forces (word commitment, attention loss, training bias, quiet errors, and “get it done” instructions) all push toward claiming success.

The Action Ledger fixes this with a simple idea: keep a runtime-managed record of what actually happened, and when something failed, force the truth into the user-facing output.

We learned this in two stages. First, the verification pass: show the model the truth and ask it to rewrite. This works most of the time. But the model still controls the rewrite, and we observed it circumventing verification through superficial compliance, narrative softening, and ledger mimicry. Asking a model to self-correct is trusting it at the exact point where you've established it can't be trusted.

The hardened version removes that trust. The runtime scrubs model-generated imitations and stamps ground-truth facts directly into the output — string concatenation, not generation. The model can narrate, but the facts are runtime-owned.

The key insight, sharpened: **the truth must come from outside the model, and it must reach the user without passing through the model.** The ledger is written by the runtime. The stamp is appended by the runtime. The model participates in the rewrite, but it does not control what the user sees about failures.

This is: - **Cheap** — zero cost when everything works, one API call when something fails - **Effective** — eliminated success hallucination in our production deployment; hardened version closes the last-mile circumvention gap - **Simple** — about 50 lines of code - **General** — works for any agent, any tools, any model

The broader principle: in any system where a probabilistic component (an LLM) reports on the outcomes of deterministic components (tool executions), the deterministic layer must own the final word. Asking the probabilistic layer to self-report accurately is hoping for the best. Stamping ground truth from the deterministic layer is engineering for the worst.

We think this will become standard. The same way assertions and contracts became standard in software engineering after Meyer, ground truth verification will become standard in agentic AI. Your agent should never hallucinate success — and the runtime should make sure it can't.

9. Availability

The Action Ledger is implemented in BRU (Bot for Routine Undertakings), an autonomous AI agent for business professionals. Code covers both cloud and local execution. Production deployment at matsyaai.com.

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